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Discrete Optimization

Matheuristics for the irregular bin packing problem with free rotations

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ABSTRACT

We present a number of variants of a constructive algorithm able to solve a wide variety of variants of the Two-Dimensional Irregular Bin Packing Problem (2DIBPP). The aim of the 2DIBPP is to pack a set of irregular pieces, which may have concavities, into stock sheets (bins) with fixed dimensions in such a way that the utilization is maximized. This problem is inspired by a real application from a ceramic company in Spain. In addition, this problem arises in other industries such as the garment industry or ship building. The constructive procedure presented in this paper allows both free orientation for the pieces, as in the case of the ceramic industry, or a finite set of orientations as in the case of the garment industry. We explicitly model the assignment of pieces to bins and compare with the more common strategy of packing bins sequentially. There are very few papers in the literature that address the bin packing problem with irregular pieces and to our knowledge this is the first to additionally consider free rotation of pieces with bin packing. We propose several Integer Programing models to determine the association between pieces and bins and then we use a Mixed Integer Programing model for placing the pieces into the bins. The computational results show that the algorithm obtains high quality results in sets of instances with different properties. We have used both industry data and the available data in the literature of 2D irregular strip packing and bin packing problems.

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1. Introduction

The paper addresses a real and highly relevant problem in industry that up to now has received minimal attention by the cutting and packing research community. This is the two-dimensional bin packing problem with irregular pieces (2DIBPP). The particular characteristics of the problem are: the shapes to be cut are irregular with concavities, the material being cut is homogenous therefore continuous rotation of the pieces are permitted, and a typical demand set of pieces require more than one stock sheet to be satisfied. While two-dimensional (2D) irregular packing problems have been extensively researched, these are almost exclusively strip packing problems. Further, only a few papers consider continuous rotation of pieces and to our knowledge there are no publications that examine problems that combine strongly heterogeneous irregular pieces, multiple stock sheets and continuous rotation. Each of these features adds complexity to the problem. While simple adaptations of known algorithms for the related simpler problem can be used, it is highly likely that these will yield sub-optimal results. Hence in this paper we seek to design an algorithm that considers and optimizes the complete problem. The particular contributions of the paper are four fold. This is the first paper to tackle the problem of bin packing irregular shapes with unrestricted rotations, and one of very few that considers pieces with concavities within the bin packing problem. We investigate some methodological design principles, in particular, the value of separating the assignment decision from the packing model versus assignment as a consequence of the packing decision. We also investigate the relative impact of assignment and packing depending on the data instance. We develop new data instances for the irregular bin packing problem based on the benchmark instances of the strip packing problem. Finally, we investigate the bias introduced in data set design with respect to the piece orientation and show how using unrestricted rotation can address this bias.

A Spanish company in the ceramic tile sector, Butech Building Technology, by PORCELANOSA Group, has developed a new solid surface which, due to its properties of strength, durability, and color stability, can be used for ventilated facades, among many other applications. The material, whose commercial name is

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Krion®, is produced in plates of 366.6 by 75 centimeters. A particular property of this product is the possibility to cut it into pieces of any polygonal shape, convex or not, according to the architect's design. This is a significant departure from the classical tiles used in ventilated facades, which are always of rectangular shape. The problem is, then, to determine how the demanded pieces are cut from the plates so as to minimize the number of required plates. As this new material is expensive and the number of pieces in a project can be very large, reducing the number of plates to a minimum can produce substantial savings. At the time of undertaking this research, the company manually designed the cutting patterns, where each order could take up to four person weeks to design the cutting pattern.

Although the present study has been motivated by this particular company, there are many other companies in which materials coming in sheets of given sizes have to be cut into smaller pieces with irregular shapes. Therefore, the ideas developed here could be applied to other situations, such as the garment industry, ship building, or the glass industry (Han, Bennell, Zhao, & Song, 2012; Martinez-Sykora, Alvarez-Valdes, Bennell, & Tamarit, 2015), maybe adapting them to the special features of each case.

The above cutting problem is a two-dimensional bin packing problem with irregular (non-convex) pieces (2DIBPP). According to the typology proposed by Wäscher, Haußner, and Schumann (2007), the 2DIBPP problem can be classified as a two-dimensional Single Bin Size Bin Packing Problem (SBSBPP) with the refinement that the pieces have irregular shape. In the case of the ceramic cutting, there are no restrictions on the angle of rotation, making the problem more difficult to solve efficiently. Further, the pieces cannot be reflected because each side of the plate has a different texture and only one side can face outwards. The objective is to minimize the total number of bins needed to cut all the pieces.

The most studied packing problem involving irregular shapes that can be found in the literature is the two-dimensional strip irregular packing problem, also known as the Nesting Problem (see Bennell and Oliveira, 2009 for a survey). There are few publications considering the bin packing problem with irregular pieces and, to our knowledge, there is no publication that additionally allows items to be continuously rotated. Chernov, Stoyan, and Romanova (2010) is one example of a number of publications by these authors that permit continuous rotation of irregular pieces. Here the objective is to minimize the area of the single containing object. An interesting application of bin packing problems with irregular pieces arises in the glass industry. In this case pieces can be placed in any orientation, however, since the cutting process forces the use of guillotine cuts, the pieces have to be convex. Han et al. (2012) propose two constructive heuristics and report results on some real data. More recently, Martinez-Sykora et al. (2015) present several MIP-based constructive procedures in which they improve the best known solutions in all the instances and, furthermore, they obtain competitive results on rectangular bin packing problems.

Terashima-Marin, Ross, Farias-Zarate, Lopez-Camacho, and Valenzuela-Rendon (2010) propose a hyper-heuristic algorithm for the 2DIBPP, which combines several placement heuristics. They combine the Bottom-Left algorithm developed by Jakobs (1996) and an improved version proposed by Liu and Teng (1999). For both procedures Terashima-Marin et al. (2010) add the possibility of rotating the pieces by a finite set of angles. Furthermore, they include different modifications of the constructive approach presented by Hifi and M'Hallah (2002). The hyper-heuristic is embedded into a Genetic Algorithm and it is tested on jigsaw puzzle instances where all the pieces are convex. Lopez-Camacho, Ochoa, Terashima-Marin, and Burke (2013) adapt the Djang and Finch heuristic (DJD), originally proposed for the one-dimension bin packing problems by Ross, Schulenburg, and Marin-Blazquez (2002), to the two-dimensional irregular bin packing problem. Despite the fact the algorithm proposed by Lopez-Camacho et al. (2013) consider non-convex pieces, in the computational experiments they only report results involving pieces with convex shapes. A later paper, (Lopez-Camacho, Terashima-Marin, Ross, & Ochoa, 2014), claim to have solved the non-convex instances, but only report the percentage of problems solved with more, the same, or less bins when compared with the problems convex counterpart. In all these approaches, the assignment of the piece to a bin is incidental to the decision of where to place the piece in the bin.

Another study that considers multiple bins and irregular pieces is that of Song and Bennell (2013), who propose a column generation procedure for solving the irregular shape cutting stock problem. The authors use the beam search algorithm proposed in Bennell and Song (2010) for generating the patterns and study three solution approaches: column generation, adapted column generation and a sequential heuristic procedure. In the computational experiments the authors build eleven instances obtained from well-known 2D irregular strip packing instances by fixing the bin length and multiplying the demand of each shape by 100. While the instances include non-convex pieces, the problem is a cutting stock problem where patterns are repeated multiple times, which is somewhat different to the two-dimensional bin packing problem considered in our study.

This paper describes a constructive algorithm that explicitly considers the two aspects of the optimization problem; the assignment of pieces to bins and the arrangement of assigned pieces in the bin. We implement several strategies for assigning pieces to the bins, including solving IP models for the one-dimensional bin packing problem and knapsack problem and heuristic methods. In a second phase, the subset of pieces assigned to a given bin are sequentially placed in the new bin using a packing procedure. This phase consists of two steps. First, we determine a set of promising rotations for the piece and then, for each rotation, we determine if the piece fits into the bin by solving a Mixed Integer Programing (MIP) model. The packing procedure permits all pieces to move within the bin to accommodate the next piece. Since the assignment is based on piece area, often the packing fails. Hence, we propose some alternative strategies for reassigning pieces. All these approaches are compared to the more common approach of deciding the bin assignment as part of the piece placement strategy.

The paper is organized as follows: in Section 2 we define the problem and some notation. In Section 3 we describe the construction algorithms that determine the assignment of pieces to bins and in Section 4 we provide the formulation for the placement of items into each bin. Section 5 contains two local search procedures designed to improve the solutions obtained. In Section 6 we describe the implementation of the constructive algorithm, including any parameters that need to be set and a full description of the data instances. In Section 7 we present the computational results and in the final section we summarize our conclusions.

2. Problem description

Let $P = \{1, \ldots, n\}$ be the set of n pieces, represented as simple polygons, to be placed into identical rectangular bins. We denote the area of piece $k \in P$ by a_k. The number of available bins can be considered large enough to pack all the pieces and the objective is to minimize the total number of bins needed to cut all the pieces in P. The width and length of the bins are denoted by W and L respectively. The pieces can be rotated continuously and must be placed entirely within the bin and may not overlap with other pieces.

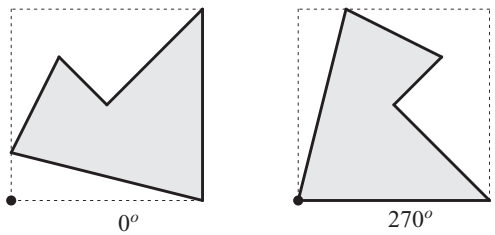


Fig. 1. Reference point of a piece in two different orientations.

A solution to the described problem is given by a set of bins $B = \{b_1, \ldots, b_N\}$. Each bin $b_i(P_i,\ O_i,\ X_i,\ Y_i)$, $\forall i = 1, \ldots, N$, is composed of a set of pieces $P_i \subseteq P$, a vector $O_i = \{o_1, \ldots, o_{n_i}\}$ defines the orientation of the pieces, where $n_i = |P_i|$ and for a given piece k, $k = 1, \ldots, n_i$, $o_k \in [0, 2\pi]$, and the two vectors, X_i and Y_i, are the X-coordinate and Y-coordinate of the reference point of each piece. We define the reference point of the piece $k \in P$ as the bottom-left corner of the enclosing rectangle of the piece when in orientation o_k. Hence, the position of the reference point, relative to the vertices of the piece, changes according to the orientation, as illustrated in Fig. 1. In addition, we consider that the bottom-left corner of each bin is placed at (0,0), so that the coordinates of the reference points are always positive. Note that, since all the pieces have to be placed, then $P = \cup_{i=1}^{N} P_i$.

The stated objective is to minimize the total number of bins. Applying this objective directly will result in many tied solutions making the solution space difficult to navigate, further in many applications, the residual material can be reused provided it is sufficiently large. Lopez-Camacho et al. (2013) propose an alternative measure of performance that maximizes the percentage usage of each bin as follows:

$$F = \frac{\sum_{i=1}^{N} U_i^2}{N} \tag{1}$$

where U_i is the utilization ratio of each bin $i \in 1 \ldots, N$, defined as:

$$U_i = \frac{\sum_{k=1}^{n_i} a_k}{LW}, \tag{2}$$

This measure avoids ties among different solutions with the same number of bins, and favors solutions with a high utilization in all but one or two bins over those that evenly distribute the pieces.

An alternative approach is used by Han et al. (2012). In order to account for the reuse of residual material, they apply a vertical or a horizontal cut to separate the non-utilized part of the bin for future use. The cut is applied to the least utilized bin only. We define the fractional number of bins as follows:

$$K = N - 1 + R^* \tag{3}$$

where R^* is the proportion of the bin used to pack pieces after applying the vertical or horizontal cut. Fig. 2 shows two different solution of the same instance (*han* taken from ESICUP benchmark instance). Although both solutions use three bins, solution (b) is better since it has a smaller value of K.

We employ both these performance measures, F and K, for comparing with the experimental results in addition to the number of bins. Also F is used to guide the local search strategies.

3. Solution algorithm

The bin packing problem has two sets of distinct decision types. Specifically these are: the assignment of pieces to a bin and how to pack the assigned pieces in the bin. Many authors design solution approaches where the assignment decision is incidental to the packing decision, for example packing a bin until it is full, closing the bin, and then opening a new bin to continue packing (Parreño, Alvarez-Valdes, Oliveira, & Tamarit, 2010). In this paper, we investigate an alternative approach where the assignment decision is solved separately and prior to determining the layout of pieces in the bin. Inevitably the assignment will often be unfeasible and one or more pieces cannot be placed in their allotted bin, hence we investigate alternative strategies for mending the assignment. In this section we describe alternative strategies for assigning to bins and mending the assignment when pieces cannot be feasibly packed. The packing procedure is described in Section 4, which packs the assigned pieces to a given bin sequentially in order of non-increasing area using a MIP model. Finally we investigate the benefit of employing local search after the construction of the solution, this is described in Section 5.

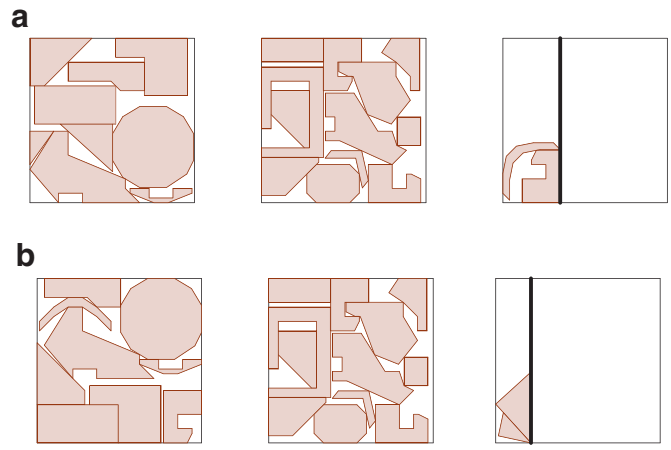


Fig. 2. Two solutions of instance *han* (Nest-MB) (a) $K = 2.35$; (b) $K = 2.20$.

The general framework for the procedure is organized as follows:

Step 1: solve a one-dimensional bin packing problem, where the pieces in P are represented by their area, in order to assign all pieces to the minimum number of bins. Sort the bins in non-increasing order of utilization area. If P is empty, stop.

Step 2: for the first bin on the list, use the MIP model described in Section 4 to arrange the assigned 2D irregular pieces into the bin. If not all the pieces fit into the bin, go to Step 3. If all the pieces can be placed in the bin with no overlap, then a feasible solution for the bin is found. Remove the bin from the list. Remove the pieces packed in the bin from P. Store the packed bin to the final solution. If all pieces are packed, stop. Otherwise, repeat Step 2 for the next bin on the list.

Step 3: if all the pieces cannot be placed into the bin, mend the assignment of pieces to the bin until a feasible arrangement is found. Remove the bin from the list. Remove pieces packed in the bin from P. Store the packed bin to the final solution. Since the original assignment has been disrupted, go to Step 1 to solve a 1D-BPP with the remaining pieces.

Sections 3.1–3.5 describe alternative procedures for solving the assignment of pieces to bins, including strategies for reassignment when the assignment is found to not be feasible. We also describe a simple construction heuristic that does not preassign pieces to bins.

3.1. Bin Packing with Greedy Decisions (BPGD)

The BPGD strategy assigns pieces to bins by solving an IP formulation of the one-dimensional bin packing problem (1DBBP) described below. Since area is a weak approximation of a sim-

ple polygon, particularly when concavities are present, the packing procedure, which decides how to arranged the 2D irregular pieces, often fails to place one or more of the assigned pieces inside the bin. While it is possible to solve the entire assignment problem again with constraints to prevent the failed bin being reproduced, there is a high chance the new assignment would result in other infeasible bins. Reaching a feasible packing through this approach would require solving many IP models with an increasing number of constraints leading to a very high computational cost. Hence a faster strategy is to retain the bin that has a feasible arrangement of some of the assigned pieces and try to improve the utilization of the bin by packing additional pieces that have been assigned to other bins. This is the approach taken by our BPGD algorithm.

The IDBBP model is defined as follows: let each piece be approximated by its area, a_i. $\overline{N}$ is an upper bound on the total number of bins required to pack all the pieces, obtained by using the First Fit Decreasing (FFD) algorithm (Johnson, Demers, Ullman, Garvey, & Graham, 1974). We define the following binary variables: s_i, $\forall i = 1, \ldots \overline{N}$, which take value 1 if bin b_i is used in the solution and 0 otherwise, and z_{ij}, $i = 1, \ldots, \overline{N}$, $j = 1, \ldots, n$, which take value 1 if piece j is assigned to bin i and 0 otherwise. An Integer Programing model for assigning pieces to bins can be formulated in a standard way as a 1DBPP as follows:

$$Min. \sum_{i=1}^{\overline{N}} s_i, \tag{4}$$

$$\sum_{j=1}^{n} a_j z_{ij} \leq LW, \;\; 1 \leq i \leq \overline{N}, \tag{5}$$

$$\sum_{i=1}^{\overline{N}} z_{ij} = 1, \;\; 1 \leq j \leq n, \tag{6}$$

$$z_{ij} \leq s_i, \;\; 1 \leq j \leq n, \;\; 1 \leq i \leq \overline{N}, \tag{7}$$

$$s_i \in \{0, 1\}, \;\; z_{ij} \in \{0, 1\} \qquad 1 \leq j \leq n, 1 \leq i \leq \overline{N}. \tag{8}$$

The objective function minimizes the total number of bins used in the solution. Constraints (5) ensure that the total area of the pieces assigned to bin b_i does not exceed the area of the bin. Equalities (6) force each piece to be assigned to one bin. Inequalities (7) guarantee that if any piece is placed into a given bin, then the corresponding bin is used in the solution. Finally, constraints (8) force the variables to be binary. Note that, if this IP model is solved to optimality and we pack all the pieces assigned to each bin successfully, the solution would be optimal.

Given the optimal solution to the IDBBP, the bins are sorted in non-increasing order of utilization. The packing procedure is applied to the first bin on this list, where the pieces assigned to the bin are sorted in non-increasing order of area. When trying to pack the pieces assigned to a bin, there are two possible outcomes:

(a) The packing procedure finds a feasible placement for all the pieces. In this case the bin and the pieces placed in the bin are removed from further consideration and saved to the final solution. The algorithm will then go on to consider the new most utilized bin on the list and apply the packing procedure.
(b) The packing procedure fails when placing a given piece, j. In this case the pieces packed in the bin so far are retained. Excluding piece j, all pieces that have not been packed (including those assigned to other bins) are sorted in non-increasing order. Next, the packing procedure tries to place each piece in turn in the partially packed bin, accepting each feasible placement. When all the pieces have been tried, the bin and the pieces placed in the bin are removed from further consideration and saved to the final solution. Since the assignment of pieces has been altered, we solve the assignment model again for the unpacked pieces.

The IP model has to be solved each time case (b) occurs because the set of remaining pieces has been modified. Hence the IP model is run at most N times. Note that each time the IP model is solved at least one bin has been closed and the number of the remaining pieces has been reduced. Therefore $\overline{N}$ is updated and the number of binary variables is reduced.

3.2. First Fit Algorithm (FF)

An alternative to solving the 1DBPP to optimality using an IP model as described above, is the use of a known heuristic algorithm. Our FF algorithm follows this strategy and adopts the fast and well known First Fit Decreasing Algorithm (FFD) (Johnson et al., 1974) to solve the 1DBPP. FFD takes an ordered list of pieces and assigns them sequentially to the bins. In order to assign a piece, FFD examines each bin in the order the bins were opened and places the piece in the first bin that can feasibly accommodate the piece. If the piece does not fit into any existing bin, a new bin is created and the piece is assigned to this new bin. This algorithm is very fast and depends critically on the initial ordering of the pieces. Therefore, in order to obtain a good solution we use the FFD assignment scheme, but starting from a unique random permutation of the pieces and repeat the process 5000 times. We keep the solution with the minimum number of bins, breaking ties by the minimum used area of the least utilized bin, and sort the bins in non-increasing order of utilization.

Similar to BPGD, the packing procedure is applied to the first bin on this list. If it finds a feasible placement for all the pieces, the bin and the pieces placed in the bin are saved to the final solution and the algorithm will go on to consider the new most utilized bin on the list.

If the packing procedure fails when placing a given piece, j, it stops and all pieces that have not been packed yet are reassigned to bins using FFD, where the partially packed bin is the first open bin and piece j cannot be assigned to it. Given the new assignment, the packing procedure tries to place the newly assigned pieces into the first bin. The packing and re-assigning continues until a feasible bin is obtained.

3.3. Partial Bin Packing (PBP)

The PBP strategy is motivated by the observation that we frequently change the assignment across the bins to obtain a feasible solution. It also recognizes that assigning across the bins prevents the algorithm being too greedy with respect to the bin under consideration. It is well documented in cutting and packing literature that a greedy solution packs small pieces together early on in the solution construction, leaving the larger more difficult pieces to the end, leading to overall poor solutions. Therefore, the PBP approach avoids reassignment by focusing on the assignment of the pieces to just one bin, but uses an objective function that favors assigning larger pieces. This strategy has been widely used in one-dimensional BPP (see Caprara and Pferschy, 2004 for a review and a worst-case analysis).

The assignment approach is a knapsack formulation, where the objective is to maximize the value of the bin by packing some of the available pieces. The following formulation sets the binary variables, q_j, $j = 1 \ldots, n$, to take the value 1 when piece j is assigned to the bin and 0 otherwise and a_{max} is the area of the biggest piece. The IP model can be formulated as follows:

$$Max. \sum_{j=1}^{n} \left(\frac{a_j}{a_{max}} \right)^2 q_j, \tag{9}$$

$$\sum_{j=1}^{n} a_j q_j \le LW, \;\; 1 \le j \le n, \tag{10}$$

$$q_j \in \{0, 1\}, \qquad 1 \le j \le n. \tag{11}$$

As with the above two strategies, the packing procedure may succeed or fail to feasibly pack the bin. If it finds a feasible placement for all the pieces, the packed bin is saved to the final solution and the algorithm will then solve the knapsack formulation with the remaining pieces to generate a new bin.

On the contrary, if the packing procedure fails when placing a given piece, j, the knapsack formulation is solved again for the remaining capacity, with j removed from consideration. Given the new assignment, the packing procedure tries to place the newly assigned pieces into the bin. The packing and re-assigning continues until a feasible bin is obtained. Piece(s) j temporarily removed from consideration are returned to the available pool of pieces and the algorithm will then solve the knapsack formulation to generate a new bin.

3.4. Two Phases Strategy (TPS)

The TPS strategy aims to enhance the BPGD strategy by attempting to distribute the small pieces across the bins during the assignment phase, as in PBP strategy. While the solutions obtained by the BPGD procedure provides the optimal number of bins required to pack all the pieces (as a 1D problem), the solution does not consider how the small and large pieces are distributed. Therefore for the TPS strategy, after solving the assignment model proposed in the BPGD strategy to find the minimum number of bins, N', we solve a second IP model to reassign pieces across bins. Let z_{ij}, $i = 1, \ldots, N', \; j = 1, \ldots, n$ be the binary variables which take value 1 if piece j is associated with bin i.

$$Min. \sum_{i=1}^{N} \sum_{j=1}^{n} \left(\frac{a_j}{a_{max}} \right)^2 \frac{2(N'-i)}{N'^2 + N'} \, z_{ij}, \tag{12}$$

$$\sum_{j=1}^{n} a_j z_{ij} \le LW, \;\; 1 \le i \le N', \tag{13}$$

$$\sum_{i=1}^{N'} z_{ij} = 1, \;\; 1 \le j \le n, \tag{14}$$

$$z_{ij} \in \{0, 1\}, \qquad 1 \le i \le N', \;\; 1 \le j \le n. \tag{15}$$

The objective function (12) forces the assignment of large pieces into the earlier bins while minimizing the total number of bins. Note that $(\frac{a_j}{a_{max}})^2$ takes a greater value when p_j has more area and $\frac{2(N'-i)}{N'^2+N'}$ takes a greater value for the earlier bins. The constraints (13) and (14) ensure, respectively, that the pieces associated to one bin does not exceed the area of the bin and that each piece is assigned to one bin.

Once the pieces have been assigned by sequentially solving both IP models, the procedure to pack and re-assign pieces is the same as the BPGD strategy.

3.5. Simple construction heuristic (SCH)

The SCH strategy is included to benchmark our approach against the commonly used approach of deciding the bin allocation and placement position together. SCH directly packs pieces into a bin in a given sorted order following a Next-Fit Decreasing strategy. The pieces are packed into the bins using the procedure described in Section 4. The structure is as follows:

(1) Sort all the pieces in order of non-decreasing area, open bin, $i = 1$.
(2) For next piece i, solve IP model described in Section 4 to pack piece i in the open bin.
(3) If a feasible solution is found, $i = i + 1$, if $i > n$, stop, else got to (2) otherwise close bin and open new bin, go to (2).

4. Packing model

In this section, we describe the model that determines the exact location and orientation of the pieces in a bin. We assume that a set of pieces $P_b \subseteq P$, ordered by non-increasing area, has been assigned to the current bin, b. The MIP model places one piece in the bin in order to minimise the weighted rectangle of the partial solution. Although, the pieces can be rotated continuously, but the MIP we use to pack the pieces requires that each piece has a fixed orientation. As a result, the MIP model is run several times for each piece solving a model for each orientation we want to consider. This idea was first proposed in Martinez-Sykora et al. (2015).

In order to obtain promising rotations for placing the next piece, we consider the edges of the pieces already placed in the bin and the edges of the bin. We calculate the angles of rotation in such a way that one edge of the piece being placed matches with either one edge of a piece already placed or one edge of the bin. If we obtain more than three different rotations, we sort the angles by the following criteria (GR):

- Non-increasing number of matches between the edges of the polygon obtained by applying the given rotation to the piece, and the edges of all the pieces already placed in the bin and the edges of the bin.
- Ties are broken by the total length of the edges of all the matchings.

If the number of rotations is not specified, we consider the first three rotations for the insertion of each piece. If in a given instance, only some fixed orientations are allowed, we use these orientations.

In what follows we give a description of the MIP model. For the sake of clarity, we first describe the model for the insertion of the first piece, then the second piece and finally the general case. The placement model for the first piece has a linear programing formulation, in which the variables x_1 and y_1 correspond to the coordinates of the reference point of piece 1:

$$Min. \; \omega L_u + (1 - \omega) W_u, \tag{16}$$

$$L_u \le L, \tag{17}$$

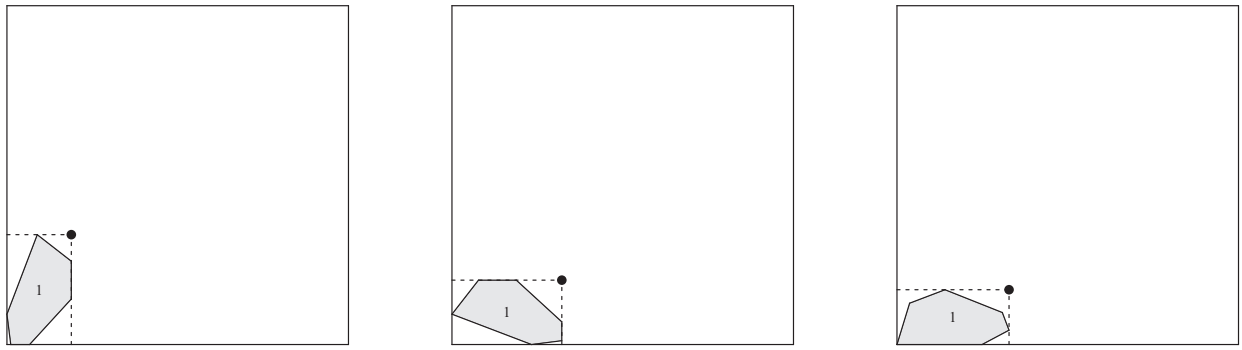
$$W_u \le W, \tag{18}$$

$$x_1 \le L_u, \tag{19}$$

$$y_1 \le W_u. \tag{20}$$

L_u and W_u denote the used length and width of the bin when placing the first piece and ω is a coefficient which depends on the dimensions of the bin. In order to balance the used length and width when adding pieces we fix $\omega = \frac{1}{(W/L)+1}$. Therefore, with this objective function the pieces will be placed in such a way that the dimensions of the enclosing rectangle are proportional to the dimensions of the bin. Constraints (17) and (18) ensure that the used length and width do not exceed the dimensions of the bin and constraints (19) and (20) force piece 1 to be inside the enclosing rectangle (L_u, W_u).

We solve this linear programing formulation once for each different orientation we consider for the piece. In this step we determine and fix the orientation of the piece for which the objective function is minimized. In Fig. 3, the first two rotations are considered because two edges of the piece are parallel to the edges



(a) Rotation 1 (b) Rotation 2 (c) Rotation 3

Fig. 3. Three rotations for the first piece (instance poly3a-x2). Rotation 3 is selected and fixed.

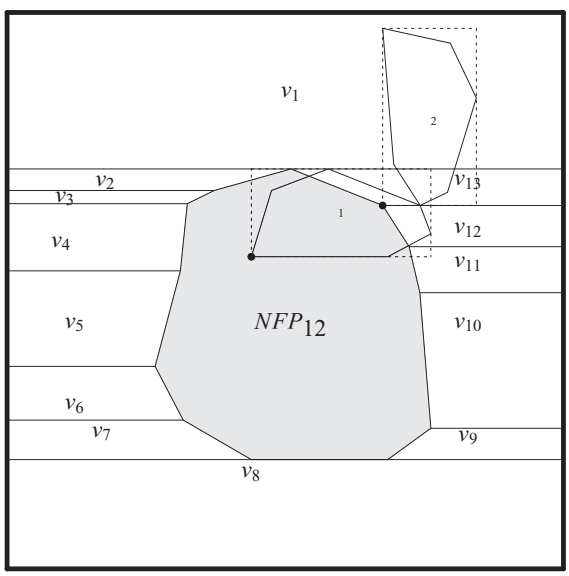


Fig. 4. NFP_{12}.

of the bin. The values of x_1 and y_1 given by the LP solution are not fixed in the following models but the orientation is. The third rotation, which happens to be the best, is considered because the largest edge of the piece is parallel to one edge of the bin.

When inserting the second and later pieces, the main difficulty is to ensure that they do not overlap. We use the *MIP*model, proposed by Alvarez-Valdes, Martinez-Sykora, and Tamarit (2013), in which non-overlapping constraints are obtained using *No-Fit Polygons (NFP)*. The *NFP* is a polygon derived from every pair of pieces in such a way that its interior represents all the overlapping positions between the pieces, and its boundary represents all the touching positions. Fig. 4 gives an example of the NFP, corresponding to pieces 1 and 2 (NFP_{12}), where the orientation of piece 1 is fixed in the previous step and piece 2 was the same shape as piece 1 and is rotated 90 degrees. Deriving the NFP is not the topic of this paper, see Bennell and Oliveira (2008) for a review of procedures. Given the correct placement of piece 1, the reference point of piece 2 must be placed outside NFP_{12} to avoid overlapping. Alvarez-Valdes et al. (2013), following the idea proposed by Fischetti and Luzzi (2008), divide the outer region into horizontal slices and define a binary variable for each slice so that it takes value 1 if the reference point is in this slice and 0 otherwise.

In the example in Fig. 4 we need 13 binary variables. The notation we use to write the formulation is a three-index notation in which the two first indexes represent the pieces, and the third index enumerates the slices (for the sake of clarity in Fig. 4 we show only the third index). In general, if pieces have complex shapes, the set of points not belonging to the NFP may have a structure more complex than that of the Fig. 4 and it might be needed to divide concave shapes into convex shapes. In these cases, before defining the slices and their associated variables, a preprocess is required and additional binary variables are associated with the subsets of the region not included in the slices (see Alvarez-Valdes et al., 2013 for a complete discussion).

Let m_{12} be the total number of binary variables, i.e, the number of slices we derived from NFP_{12} ($m_{12} = 13$ in Fig. 4). We denote by t_{12i} the total number of inequalities required to describe slice S^i associated with v_{12i}, $i = 1, \ldots, m_{12}$. Despite the fact that in Fig. 4 $t_{12i} = 4$, $\forall i = 1, \ldots, m_{12}$, it might be possible that t_{12i} takes different values if the NFP_{12} has a more complex shape. In what follows we describe first the non-overlapping constraints and then the containment constraints.

If, for some slice k of the NFP_{12}, $v_{12k} = 1$, the constraints defined by the sides of the slice:

$$\alpha_{12}^{kf}(x_2 - x_1) + \beta_{12}^{kf}(y_2 - y_1) \le \delta_{12}^{kf} \qquad f = 1, \ldots, t_{12k} \tag{21}$$

where α_{12}^{kf} and β_{12}^{kf} are the coefficients defining the slope of the inequality and δ_{12}^{kf} the intercept, must be satisfied. However, if $v_{12k} = 0$, these constraints must be relaxed. This can be done by using big-M constants:

$$\alpha_{12}^{kf}(x_2 - x_1) + \beta_{12}^{kf}(y_2 - y_1) \le \delta_{12}^{kf} + (1 - v_{12k})M \quad f = 1, \ldots, t_{12k} \tag{22}$$

Fischetti and Luzzi (2008) describe a procedure for building these constraints avoiding the big-M constants. Since exactly one slice is used in any feasible solution, this equality holds:

$$\sum_{i=1}^{m_{12}} v_{12i} = 1 \tag{23}$$

Then a tighter constant could be written for each slice, eliminating the big-M and including on the right hand side of the inequality all the binary variables, defined by NFP_{12}, multiplied by a given constant. These constants are obtained by solving the following problem:

$$\delta_{12}^{kfh} = max_{p_2 \in S^h}\ \alpha_{12}^{kf}(x_2 - x_1) + \beta_{12}^{kf}(y_2 - y_1) \tag{24}$$

which corresponds to the maximum value of the left hand side when piece 2 lies on slice h, $h \in \{1, \ldots, m_{12}\}$.

Then, the non-overlapping inequalities can be written as follows:

$$\alpha_{12}^{kf}(x_2 - x_1) + \beta_{12}^{kf}(y_2 - y_1) \le \sum_{h=1}^{m_{12}} \delta_{12}^{kfh} v_{12h}$$
$$k = 1, \ldots, m_{12}, \;\; f = 1, \ldots, t_{12k} \tag{25}$$

The containment constraints ensure that each piece does not exceed the limits of the bin (as with inequalities (19) and (20) in the formulation for one piece). Alvarez-Valdes et al. (2013) describe a lifting procedure for these inequalities, adding some of the binary variables of the corresponding *NFP*, associated with slices on which one piece protrudes either horizontally or vertically from the other. Let us denote by $\bar{y}_{12k}$ and $\underline{y}_{12k}$ ($\bar{x}_{12k}$ and $\underline{x}_{12k}$) the maximum and minimum value of $y_2 - y_1$ (respectively, $x_2 - x_1$), when slice k is used. Let $\underline{X}^{12}$ and $\underline{Y}^{12}$ be the minimum X-coordinate and Y-coordinate value of NFP_{12} respectively. We define the following classification of the binary variables:

- $U_{12} := \{v_{12k} \mid \underline{y}_{12k} \ge 0\}$, the set of binary variables whose associated slices do not allow piece 2 to protrude from below piece 1. In Fig. 4, $U_{12} = \{v_1, v_2, v_3, v_{12}\}$.
- $R_{12} := \{v_{12k} \mid \underline{x}_{12k} \ge 0\}$, the set of binary variables whose associated slices do not allow piece 2 to protrude from the left of piece 1. In Fig. 4, $R_{12} = \{v_9, v_{10}, v_{11}, v_{12}\}$.
- $LS_{12} := \{v_{12k} \mid \lambda_{12k} > 0\}$ where $\lambda_{12k} = l_1 - (\bar{x}_{12k} - \underline{X}^{12})$ and l_1 is the length of piece 1 in the current orientation. Then, LS_{12} is the set of binary variables whose associated slices force piece 1 to protrude from the right of piece 2. In Fig. 4, $LS_{12} = \{v_2, v_3, v_4, v_5, v_6, v_7\}$.
- $DS_{12} := \{v_{12k} \mid \mu_{12k} > 0\}$ where $\mu_{12k} = w_1 - (\bar{y}_{12k} - \underline{Y}^{12})$ and w_1 denotes the width of piece 1. Then, DS_{12} is the set of binary variables whose associated slices force piece 1 to protrude from above piece 2. In Fig. 4, $DS_{12} = \{v_7, v_8, v_9\}$, variable v_6 is not in DS_{12} because $\mu_{126} = 0$.

The containment constraints for piece 1 are:

$$\sum_{k \in R_{12}} \underline{x}_{12k} v_{12k} \le x_1 \le L_u - l_1 - \sum_{k \in LS_{12}} \lambda_{12k} v_{12k} \tag{26}$$

$$\sum_{k \in U_{12}} \underline{y}_{12k} v_{12k} \le y_1 \le W_u - w_1 - \sum_{k \in DS_{12}} \mu_{12k} v_{12k} \tag{27}$$

Similarly we can build the containment constraints for piece 2. In Fig. 4, $U_{21} = \{v_5, v_6, v_7, v_8, v_9, v_{10}\}$, $R_{21} = \{v_2, v_3, v_4, v_5, v_6, v_7\}$, $LS_{21} = \{v_9, v_{10}, v_{11}, v_{12}\}$, and $DS_{21} = \{v_1, v_2, v_3, v_4, v_{11}, v_{12}\}$. The containment constraints for piece 2 can be written as follows:

$$\sum_{k \in R_{21}} \lambda_{21k} v_{12k} \le x_2 \le L_u - l_2 - \sum_{k \in LS_{21}} \underline{x}_{12k} v_{12k} \tag{28}$$

$$\sum_{k \in U_{21}} \bar{y}_{12k} v_{12k} \le y_1 \le W_u - w_2 - \sum_{k \in DS_{21}} \mu_{21k} v_{12k} \tag{29}$$

where $\lambda_{21k} = l_2 - (\bar{X}^{12} - \underline{x}_{12k})$ and $\mu_{21k} = w_2 - (\bar{Y}^{12} - \underline{y}_{12k})$.

The MIP formulation we consider to place pieces 1 and 2 has as objective function expression (16), subject to:

- Constraints (17) and (18) to ensure that L_u and W_u do not exceed the limits of the bin.
- Lifted containment constraints (26) and (27) for piece 1.
- Lifted containment constraints (28) and (29) for piece 2.
- Non-overlapping constraints (23) and (25).

This MIP model is solved three times since we consider three orientations as illustrated in Fig. 5. Each time we solve the model for a given rotation and we obtain an improved feasible solution, the value of its objective function is used as an upper bound for the following rotations. Therefore, only models producing improved solutions are solved to optimality. If the orientation being tested cannot produce a better solution, the model is quickly identified as unfeasible by the solver, speeding up the process. In the three solutions depicted in Fig. 5 the orientation of piece 1 is fixed, as a result of the previous step, but its position is determined by the solution of the current model.

There are 13 binary variables in each one of the models solved for the insertion of the second piece. Since this number increases exponentially with the number of pieces already placed into the bin, it maybe necessary to fix the relative position between the pieces already placed to reduce this number. If the computational time needed to solve the MIP model to optimality exceeds a given threshold, θ, then for the following insertion we fix the binary variables to the values obtained in the previous model. Otherwise, we allow more flexibility by keeping the binary variables free. Further each MIP model is given a maximum computation time.

For the general case, where the next piece to be inserted is piece j, $j-1$ pieces have been inserted and now have fixed orientation. The model for the insertion of j can be written as follows:

$$Min.\ \omega L_u + (1-\omega) W_u, \tag{30}$$

$$L_u \le L, \tag{31}$$

$$W_u \le W, \tag{32}$$

$$\sum_{k \in R} \underline{x}_{j_1 j_2 k} v_{j_1 j_2 k} \le x_{j_1} \le L_u - l_{j_1} - \sum_{k \in LS_{j_1 j_2}} \lambda_{j_1 j_2 k} v_{12k},$$
$$j_1, j_2 \in \{1, \ldots, j\} \tag{33}$$

$$\sum_{k \in U_{j_1 j_2}} \underline{y}_{j_1 j_2 k} v_{j_1 j_2 k} \le y_{j_1} \le W_u - w_{j_1} - \sum_{k \in DS_{j_1 j_2}} \mu_{j_1 j_2 k} v_{j_1 j_2 k},$$
$$j_1, j_2 \in \{1, \ldots, j\} \tag{34}$$

$$\alpha_{j_1 j_2}^{kf}(x_{j_2} - x_{j_1}) + \beta_{j_1 j_2}^{kf}(y_{j_2} - y_{j_1}) \le \sum_{h=1}^{m_{j_1 j_2}} \delta_{j_1 j_2}^{kfh} v_{j_1 j_2 h},$$
$$1 \le j_1 < j_2 \le j, k = 1, \ldots, m_{j_1 j_2}, f = 1, \ldots, t_{j_1 j_2 k} \tag{35}$$

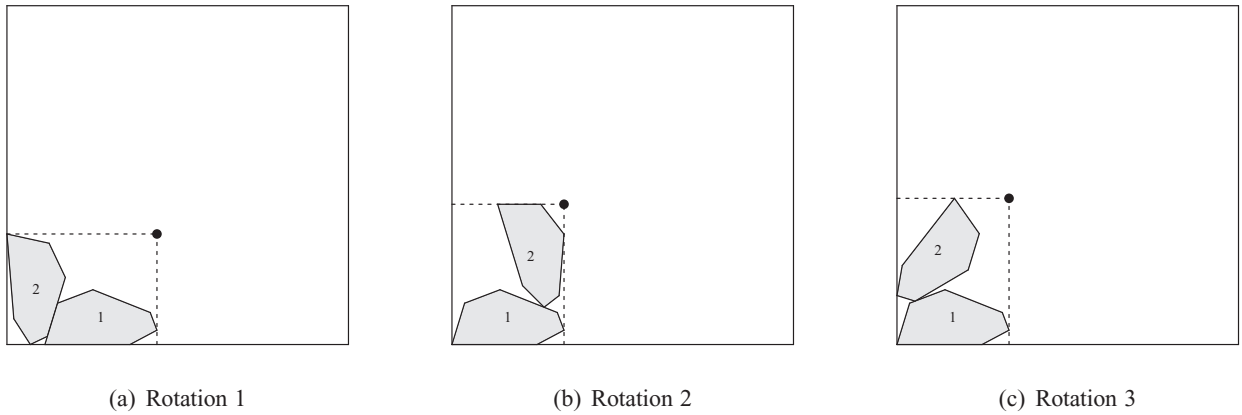
$$\sum_{i=1}^{m_{j_1 j_2}} v_{12i} = 1, \quad 1 \le j_1 < j_2 \le i \tag{36}$$

$$v_{j_1 j_2 k} \in \{0, 1\}, \quad j_1, j_2 \in \{1, \ldots, j\} \tag{37}$$

$$x_l, \; y_l \ge 0, \quad 1 \le l \le j \tag{38}$$

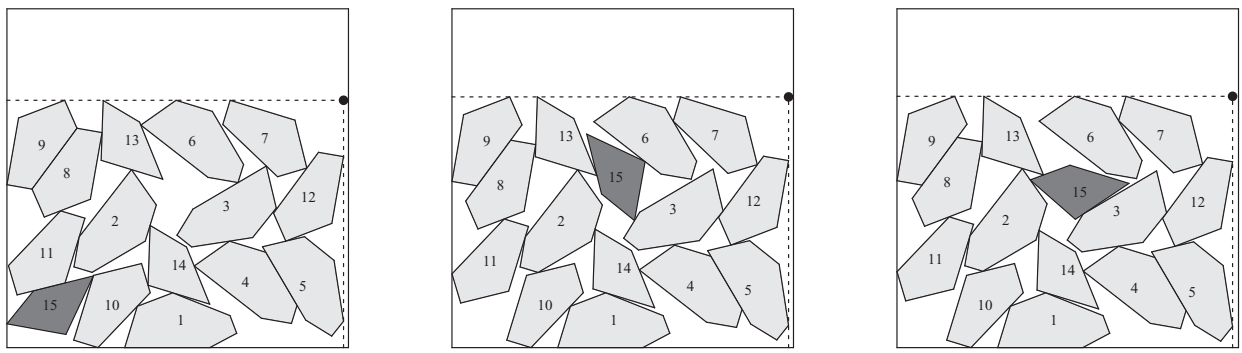
Constraints (31) and (32) ensure that L_u and W_u do not exceed the dimensions of the bin. The containment constraints (33) and (34) are required for all the pairs of pieces taking into account the order ($j(j-1)$ pairs) and the non-overlapping constraints (35) and (36) are required for all the NFP ($\frac{j(j-1)}{2}$ pairs).

In Fig. 6 we show the insertion of piece 15. In this example the previous MIP model was difficult to solve to optimality (the computation time was greater that the threshold θ). Therefore, the relative position of the pieces already placed is fixed while piece 15 is being inserted. We can see that we obtain different solutions with the same objective function value. In this case, the algorithm would solve the first MIP model to optimality (Rotation 1) obtaining a solution. This objective function value will be used in the next two MIPs as upper bound. As a result, the second and third model will be quickly identified as infeasible. Their solutions appear in Fig. 6 only for illustrative purposes.



(a) Rotation 1 (b) Rotation 2 (c) Rotation 3

Fig. 5. Three rotations for the second piece. Rotation 3 is selected and fixed.



(a) Rotation 1 (b) Rotation 2 (c) Rotation 3

Fig. 6. Three rotations for the 15th piece. Rotation 1 is selected and fixed.

5. Local search

In this section we present two local search strategies referred to as *LS1* and *LS2* respectively. Throughout the different proposed methods we consider that a solution has been improved if its coefficient F (Eq. (1)) increases. Both strategies are based on hill climbing first found improvement and do not include diversification schemes. Both terminate when a local optimum solution is found. The rationale behind such simplistic procedures is that coefficient F has several properties: First, if a move leads to a reduction on the number of bins, F increases and the move is accepted. Second, if a move results in the same number of bins but with one bin with a higher utilization, F will still increase, accepting the move. These properties prove useful when differentiating between solutions with the same number of bins, removing large plateaux from the solution space.

5.1. LS1

Given a solution containing a set of bins $B = \{b_1, \ldots, b_N\}$ sorted by non-decreasing utilization, i.e, $U_1 \leq U_2 \leq \cdots \leq U_N$. The proposed local search procedure (*LS1*) considers all pairs of bins, b_i and b_j with $i < j$, and attempts to move pieces one by one from bin i into bin j. If all pieces from both bins fit into only one bin, the new solutions is accepted. Otherwise, the move is accepted if U_j is increased and consequently U_i is decreased. The new solution would have the same number of bins but a higher F. The idea is to reduce the usage of the least used bins so that eventually they might be emptied and consequently removed.

A neighborhood move involves the removal of one piece from the fullest bin, b_j, starting with the smallest piece in the bin, and the insertions of a piece from emptiest bin, b_i, starting from the largest piece in the bin. Each piece from b_i is tried before the piece from b_j is returned and the next smallest piece is removed. The first improving move found is accepted.

The neighborhood search procedure *LS1* is as follows: we first sort the pieces from bin b_j by non-decreasing area and the pieces from bin b_i by non-increasing area. Considering the pieces from each bin in their sorted order, remove a piece $p_j \in b_j$. This generates an empty space in which all pieces $p_i \in b_i$ are considered. Each piece from b_i is packed into b_j, one at a time. If it fits, the piece is kept in b_j and the next piece in b_i is considered. If not, the piece is placed onto an *Unplaced* list of pieces. When all pieces of b_i have been considered, we then attempt to pack the removed piece p_j into b_j. Again, if it does not fit, it is put onto the *Unplaced* list.

At this stage, we examine the new utilization of bin b_j. If it has not improved, then the move is discarded. Otherwise, if *Unplaced* is empty, then bin b_i is also empty and the new solution is accepted. If *Unplaced* is not empty, all of its pieces are packed into a new bin, again one at a time in non-increasing order of their area. If this packing is unsuccessful, then the new solution has a larger number of bins and is discarded. If all pieces in *Unplaced* fit, the solution is accepted if the recalculated F is higher. The complete procedure is further detailed in Algorithm 1.

LS1 considers each pair of bins starting with those with the lowest utilization. Each time an improvement is found we start again from the new solution studying all pairs of bins in which

Algorithm 1: Procedure to explore the neighborhood in *LS1*.

```
Data: b_i and b_j such that u_i ≤ u_j
Result: b_i' and b_j' such that u_i' ≤ u_i and u_j' ≥ u_j
b_i' = b_i;
b_j' = b_j;
for each piece p_j in b_j do
    Set area_out = area_{p_j}, area_in = 0, Unplaced = ∅;
    Remove p_j from b_j';
    List: Sort pieces in b_i by non-increasing area;
    Add p_j to the end of List;
    for each piece p in List do
        Obtain rotations piece p described in Section 4;
        Solve the corresponding MIP models (one per
        rotation);
        if piece p fits then
            Update b_j';
            area_in += area_p;
        else
            Unplaced = Unplaced ∪ {p};
        end
    end
    if area_in > area_out (bin b_j' has a higher utilization) then
        if Unplaced ≠ ∅ then
            Sort Unplaced by non-increasing area;
            Build b_i' trying to place all the pieces into the bin
            (see Section 4);
            if All pieces fit then
                return b_i' and b_j';
            else
                Set b_i' = b_i and b_j' = b_j;
            end
        else
            return b_j' and b_i' as an empty set (all the pieces fits
            only in one bin);
        end
    else
        Set b_i' = b_i and b_j' = b_j;
    end
end
```

exactly one bin is different from the previous solution and all previously unvisited pairs of bins.

5.2. LS2

This local search aims to extend the scope of the search in *LS1* by considering a set of bins, B', instead of a single bin b_j. As a result, we increase the chances of packing all pieces of bin b_i into a set of bins. As before, the aim is to empty bin b_i, as much as possible, from a solution with N bins by moving pieces into any of the bins with greater utilization than b_i. The set of bins, B' is all the bins with greater utilization that b_i and lower than 0.99, say $B' = \{b_{j_1}, \ldots, b_{j_r}\}, r < N, B' \subseteq B \setminus \{b_i\}$, where r is the number of bins in B'.

The main difference is that in *LS1*, once we have built a new bin b'_j, we try to rebuild b_i. Whereas in *LS2*, we consider $r \geq 2$ bins in such a way that bin b_i is not rebuilt until all the r bins have been considered. If we fail to pack all the pieces from the list of unpacked pieces in the new bin b'_i, the whole movement fails and we return to the previous solution. The specific details of the procedure are given as pseudocode in Algorithm 2.

Algorithm 2: Procedure to explore the neighborhood in *LS2*.

```
Data: b_i, b_{j_1}, ..., b_{j_r} such that u_i ≤ u_{j_1} ≤ ··· ≤ u_{j_r}
Result: r or r+1 new bins packing the same set of pieces
        with better coefficient F
Copy pieces of b_i in List;
b_i = ∅;
Copy bins b'_{j_1} = b_{j_1}, ..., b'_{j_r} = b_{j_r};
improved = true;
while improved do
    improved=false;
    for k = 1 to r do
        for each piece p_{j_k} in b'_{j_k} do
            Set area_out = area_{p_{j_k}}, area_in = 0;
            Remove p_{j_k} from b'_{j_k};
            Copy List2 = List;
            Add p_{j_k} to the end of List;
            for each piece p in List2 do
                Obtain rotations for piece p as described in
                Section 4;
                Solve the corresponding MIP models (one per
                rotation);
                if piece p fits then
                    Update b'_{j_k};
                    area_in += area_p;
                    Remove p from List2;
                else
                    Study next piece;
                end
            end
            if area_in > area_out then
                if List2 ≠ ∅ then
                    List = List2 and sort the list by
                    non-increasing area;
                    b_{j_k} = b'_{j_k};
                    improved=true;
                else
                    return b'_{j_1}, ..., b'_{j_r} (a solution with one bin
                    less have been found);
                end
            else
                b'_{j_k} = b_{j_k};
            end
        end
    end
    if List has been modified then
        Try to place all the pieces from List into b_i, building a
        new bin b_i';
        if All pieces fit then
            Improvement is found: return b_i and bins
            b'_{j_1}, ..., b'_{j_r};
        else
            return (no improvement is found);
        end
    else
        return (no improvement is found);
    end
end
```

Local search *LS2* is based on applying the procedure described in Algorithm 2, which is carried out for all the bins b_i, $i = 1, \ldots, N-1$. Note that i goes all the way up to the second fullest bin without taking into account bins with an utilization greater than 0.99.

6. Implementation

In Section 3 we present five different construction algorithms for solving the 2DIBBP: BPGD, FFD, PBP, TPS and SCH. The first four approaches initially solve a variant of the one dimensional bin packing problem prior to packing the 2D pieces in the bins. The fifth approach performs the assignment and the packing together. All approaches use the MIP model described in Section 4 to pack the pieces.

In Section 5 we described two local search procedures. Although they can be applied to the solutions obtained by any constructive algorithm, we have tested them only in combination with the constructive algorithm *PBP*. This is because PBP produces more efficient layouts at lower computational cost on average than the other variants (see Section 7).

There are few parameters to set. The construction algorithms terminate when they find a feasible solution and the local search strategies terminate when they hit a local optima. As described in Section 4, the MIP models can be time consuming, hence for all the approaches we consider $\theta = 20$, so if the time required to solve the MIP exceeds 20 seconds, the binary variables are fixed in the next insertion. We also fix a time limit of 50 seconds per MIP solved. Our experience in developing the algorithms was that this gave a reasonable tradeoff between computation time and solution efficiency. Clearly if the MIP model is permitted to run for longer, better solutions can be found. The only other parameter is related to the number of rotations attempted in the packing model. For instances in which the pieces can rotate freely, their orientation will be obtained by the criteria described in Section 4. We denote by GR1 the criterion in which only the best rotation is considered and by GR3 the case in which the best three rotations are studied. For instances in the literature in which the orientations of the pieces are limited, usually to two angles of rotation (0 degree, 180 degrees), or four angles of rotation (0 degree, 90 degrees, 180 degrees, 270 degrees), we consider only these allowed orientations.

The five construction algorithms and the two local search strategies are coded in C++ and implemented using Visual Studio 2008. We use *CPLEX*, version 12.6.0.0, for solving both the Integer Programing models in the assignment strategy and the Mixed Integer Programing models in the packing strategy, and the computational tests are run on a PC with core i7 2600 processor and 4 gigabytes memory.

6.1. data

The different variants of the algorithm were tested on four different sets of instances with irregular shapes available in the literature. The first two sets of instances were proposed by Lopez-Camacho et al. (2013). Both sets are jigsaw puzzle instances where the optimal solution is known and equal to 100% utilization of every bin. The first set, JP1, has a collection of 540 instances where all pieces are convex. These instances are divided in 18 classes with 30 problems in each class with varying numbers of pieces per bin. The second set, JP2, has 480 instances (16 classes with 30 problems each class) in which the pieces have concavities. These two sets of instances are available online at ESICUP web site (http://paginas.fe.up.pt/esicup/tiki-index.php).

Fig. 7 shows the solution to two JP1 instances of different types. The size of the bins is the same (but scaled differently in the picture). However, the shape of the pieces and the number of pieces per bin are very different. Both solutions are obtained by algorithm PBP. The top solution belongs to an instance of class L and the solution depicted at the bottom is an instance of class O (in which the algorithm was able to find the optimal solution).

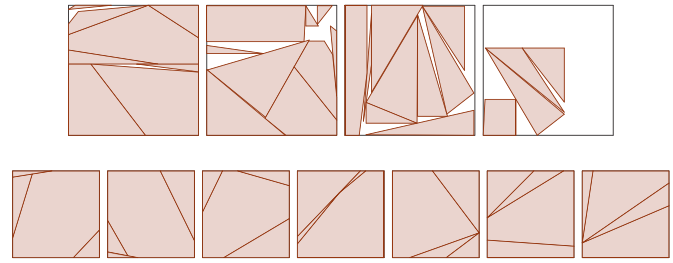


Fig. 7. Two solutions of different type instances.

Table 1
Irregular strip packing instances.

	N	SB	MB	LB	Rot
albano	24	3337.40	4551.00	6068.00	0–180
shapes2	28	5.50	7.50	10.00	0–180
trousers	64	64.90	88.50	118.00	0–180
shapes0	43	15.40	21.00	28.00	0–90–180–270
shapes1	43	15.40	21.00	28.00	0–180
shirts	99	26.00	19.50	14.30	0–180
dighe2	10	77.00	105.00	140.00	0
dighe1	16	72.60	99.00	132.00	0
fu	12	15.40	21.00	28.00	0–90–180–270
han	23	25.30	34.50	46.00	0–90–180–270
jakobs1	25	8.80	12.00	16.00	0–90–180–270
jakobs2	25	17.60	24.00	32.00	0–90–180–270
mao	20	1206.70	1645.50	2194.00	0–180
poly1a	15	14.30	19.50	26.00	0–90–180–270
poly2a	30	14.30	19.50	26.00	0–90–180–270
poly3a	45	14.30	19.50	26.00	0–90–180–270
poly4a	60	14.30	19.50	26.00	0–90–180–270
poly5a	75	14.30	19.50	26.00	0–90–180–270
poly2b	30	14.30	19.50	26.00	0–90–180–270
poly3b	45	14.30	19.50	26.00	0–90–180–270
poly4b	60	14.30	19.50	26.00	0–90–180–270
poly5b	75	14.30	19.50	26.00	0–90–180–270
swimm	48	2133.61	2909.47	3879.29	0–180

The third set of instances are created from the well-known irregular strip packing benchmark instances, available on the ESICUP web site. These have a high variety of shapes (convex and non-convex) and the pieces do not fit together exactly, as in the previous two sets. In strip packing problems only the width of the stock sheet is constrained, and the objective is to minimize the total length required to pack all the pieces. In order to modify these instances to our bin packing problem, we only need to define a fixed stock sheet size (width and length). We have built three sets of instances according to the bin size as follows. Let m_d be the maximum length or width across all the pieces in their initial orientation for a given instance. Then the bin are a fixed dimensional square with the following sizes (given in the same units of the pieces):

- *Nest-SB* (small bins). The bin dimensions are $W = L = 1.1m_d$.
- *Nest-MB* (medium bins). The bin dimensions are $W = L = 1.5m_d$.
- *Nest-LB* (large bins). The bin dimensions are $W = L = 2m_d$.

There are twenty-three irregular strip packing instances, which gives sixty-nine bin packing instances arizing from the three bin sizes. Table 1 shows the instances with the corresponding number of pieces, bin size, and rotations permitted. Columns SB, MB and LB shows the size of the bins in each one of the sets.

Table 2
Comparison between different variants of the constructive algorithm.

	SCH			BPGD			FF			PBP			TPS		
	N	*F*	*K*	*N*	*F*	*K*	*N*	*F*	*K*	*N*	*F*	*K*	*N*	*F*	*K*
JP1	7.600	0.682	7.189	7.670	0.671	7.289	7.922	0.632	7.521	7.569	0.685	7.159	7.656	0.672	7.275
JP2	7.204	0.693	6.834	7.340	0.676	7.007	7.633	0.629	7.282	7.159	0.699	6.805	7.308	0.680	6.981
Nest-SB	9.174	0.409	8.755	9.261	0.395	8.884	9.304	0.392	8.913	9.174	0.409	8.737	9.348	0.396	8.938
Nest-MB	4.870	0.425	4.476	4.870	0.425	4.458	4.870	0.427	4.464	4.870	0.425	4.477	4.870	0.425	4.485
Nest-LB	2.783	0.398	2.386	2.783	0.398	2.395	2.783	0.398	2.396	2.783	0.398	2.381	2.826	0.391	2.413
Jotika	13.125	0.695	12.542	13.375	0.678	12.790	13.500	0.664	12.892	13.125	0.698	12.514	13.375	0.671	12.884
Av.	7.459	0.550	7.030	7.550	0.540	7.137	7.669	0.524	7.245	7.447	0.553	7.012	7.564	0.539	7.163

Table 3
Computational time (in seconds) used by all the constructive algorithms.

	SCH	BPGD	FF	PBP	TPS
JP1	52	81	54	53	85
JP2	72	58	40	42	67
Nest-SB	78	67	61	72	98
Nest-MB	129	119	121	116	124
Nest-LB	291	313	284	265	307
Jotika	129	163	154	148	233
Av.	125	134	120	116	152

Table 4
LS1 vs LS2.

	PBP		LS1			LS2		
	F	*T*	*F*	%Imp	*T*	*F*	%Imp	*T*
JP1	0.685	52	0.703	1.740	129	0.723	**3.806**	161
JP2	0.699	42	0.716	1.753	136	0.729	**3.068**	123
Nest-SB	0.409	72	0.434	**5.802**	1721	0.432	5.389	1297
Nest-MB	0.425	116	0.445	4.493	2816	0.452	**6.029**	3172
Nest-LB	0.398	265	0.403	1.222	951	0.403	**1.294**	767
Jotika	0.698	148	0.725	3.740	1931	0.730	**4.370**	1426

Table 5
Analysis of different rotation criteria on Nest-MB instances.

	Initial Random		Initial Given		Free Rot	
	N	*F*	*N*	*F*	*N*	*F*
albano	4	0.332	3	0.480	3	**0.510**
shapes2	12	0.248	10	0.349	10	**0.350**
trousers	4	0.362	3	0.555	3	**0.580**
shirts	9	0.450	8	0.518	8	**0.570**
shapes0	10	0.179	7	0.271	6	**0.390**
shapes1	9	0.150	7	0.282	6	**0.390**
dighe2	2	0.236	1	**0.823**	2	0.280
dighe1	2	0.283	2	**0.368**	2	0.290
fu	5	0.247	4	**0.443**	4	0.440
han	3	0.309	3	0.387	3	**0.390**
jakobs1	6	0.281	4	**0.570**	4	0.530
jakobs2	4	0.354	4	**0.401**	4	0.380
mao	3	0.272	3	0.271	3	**0.300**
poly1a	2	0.307	2	0.308	2	**0.320**
poly2a	4	**0.342**	4	0.339	4	0.340
poly3a	5	0.420	5	0.419	5	**0.420**
poly4a	7	0.396	7	0.392	7	**0.410**
poly5a	8	0.448	8	0.449	8	**0.450**
poly2b	4	0.363	4	0.389	4	**0.400**
poly3b	5	0.423	5	0.432	5	**0.440**
poly4b	6	**0.474**	6	0.465	6	0.470
poly5b	7	**0.486**	7	0.478	7	0.480
swimm	5	0.385	5	**0.397**	5	0.390
Av.	5.478	0.337	4.870	0.425	4.826	0.414
#best		3		6		14

The rationale for considering three different bin sizes is to investigate the relative importance of the assignment and the packing approaches. Intuitively, for instances *Nest-SB* the assignment problem should be more important than for instances *Nest-MB* and *Nest-LB* because fewer pieces can fit into one bin and the packing problem will be easier. Conversely, in the instances of *Nest-LB*, the packing phase will be more important, where solving the packing problem more efficiently may lead to better solutions even if the assignment is not ideal.

The fourth set of instances have been obtained from Han et al. (2012) for the 2-Dimensional Bin Packing Problem with Irregular convex Pieces and Guillotine Cuts (2DBPPIPGC). The authors test their algorithms on 8 instances. 4 of these instances, called J40, J50, J60 and J70, are real data obtained by a company in the glass cutting industry. The remaining 4 instances, called H80, H100, H120 and H149, were generated randomly taking into account the main properties of the real data. This data set permits free rotation of the pieces. We denote these 8 instances by Jotika.

In addition, we report the solutions we obtain for the three real instances for the ceramic industry provided by a Spanish company located in Castellon, BUTECH-PORCELANOSA. The instances have 209, 340 and 484 pieces. Two of the instances have pieces with concavities and one instance has only convex pieces. In this real application there is no restriction on the orientation of the pieces, and the objective is to minimize the total number of bins (N).

7. Computational results

In the first computational test we compare the five variants of the constructive algorithm presented in Section 3. In Table 2 we present the average results obtained by the algorithms in all instances in each data set. For each one of the variants we report the average number of bins (N), the average value of F described in (1) and the average fractional number of bins K defined in (3). Table 3 shows the average computational time (T) in seconds.

Although the differences in N, the number of bins required by the solutions, are very small, as is usual in bin packing problems, we can observe that in all the data sets PBP performed the best on average in all the measures of performance (N, F and K), with lower computational effort. Variant SCH is the next best preforming algorithm, although the computational effort is slightly higher than PBP and FF. The approaches that made a full assignment to the bins performed least well (BPGD, FF and TPS). PBP makes a partial assignment by solving the 1D knapsack problem for one bin at a time and reducing the greediness of the approach with a modified objective function. SCH does not pre-assign pieces to bins but simply places the piece in the first bin it will fit. This leads to many more attempts to pack a piece, increasing computation time. Hence, we can conclude that attempting to assign pieces prior to packing is valuable but only for the immediate decision to pack the next bin. Tables 7–12 in Appendix A shows the average results for each one of the classes (or individual instances) in JP1, JP2, Nest and Jotika. Since *PBP* is the best performing algorithm, we use this variant for the remainder of the investigation.

Table 6
Computational results for the real data instances.

	SCH			BPGD			PBP			FFD		
	N	*U*	*T*	*N*	*U*	*T*	*N*	*U*	*T*	*N*	*U*	*T*
BP209	**48**	0.9157	320	52	0.8454	1280	**48**	0.9157	293	51	0.8577	589
BP340	204	0.9354	663	214	0.8918	4049	**202**	0.9455	234	210	0.9089	568
BP484	**316**	0.9732	22	388	0.7900	1827	**316**	0.9749	83	325	0.9471	527

Table 7
Results of the different constructive algorithms applied on set of instances JP1.

	SCH				BPDG				FF				PBP				TPS			
	N	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*
A	4.000	0.613	3.498	51.733	4.000	0.605	3.520	61.400	4.000	0.605	3.519	47.600	4.000	**0.614**	3.490	51.700	4	0.605	3.5356	61.1
B	11.833	0.761	11.406	0.933	11.333	**0.826**	11.100	27.833	11.967	0.741	11.586	1.733	11.700	0.781	11.281	1.300	11.5	0.8059	11.243	24.3
C	7.233	**0.731**	6.838	18.200	7.600	0.671	7.155	46.267	7.933	0.615	7.436	22.933	7.233	0.729	6.858	18.833	7.6	0.671	7.1642	46.567
D	4.000	**0.591**	3.638	149.500	4.000	0.586	3.668	160.340	4.000	0.581	3.693	163.700	4.000	0.590	3.636	164.533	4	0.5856	3.667	168.73
E	4.400	**0.523**	4.098	100.500	4.400	0.519	4.102	127.367	4.467	0.510	4.141	98.300	4.400	0.522	4.098	101.167	4.5667	0.4961	4.1762	109.4
F	3.000	**0.516**	2.412	39.733	3.000	0.508	2.457	31.800	3.000	0.507	2.435	34.900	3.000	0.515	2.414	41.967	3	0.5041	2.4467	34.4
G	14.533	0.716	14.158	1.833	14.867	0.684	14.494	78.100	14.967	0.681	14.552	3.233	14.467	**0.724**	14.050	3.300	14.633	0.7082	14.272	68.967
H	14.267	0.751	13.859	1.100	14.067	0.774	13.719	68.133	14.400	0.736	14.037	2.033	14.133	0.766	13.693	1.833	14	**0.786**	13.693	66.267
I	4.000	**0.629**	3.383	87.067	4.000	0.622	3.410	97.300	4.000	0.617	3.414	93.233	4.000	0.628	3.382	83.367	4	0.6211	3.4057	97.733
J	5.000	0.671	4.578	88.233	5.000	0.665	4.619	97.033	5.000	0.662	4.628	82.100	5.000	**0.672**	4.570	90.533	5	0.6656	4.6287	94.767
K	7.033	0.760	6.730	47.800	7.233	0.728	6.896	72.733	7.567	0.677	7.184	37.733	7.033	**0.761**	6.721	51.533	7.2333	0.7274	6.9074	73.2
L	4.067	**0.589**	3.751	28.467	4.067	0.580	3.806	49.367	4.100	0.569	3.856	34.867	4.067	0.588	3.754	28.600	4.0667	0.5782	3.8204	44.5
M	6.400	**0.658**	6.102	24.200	6.500	0.636	6.198	48.400	7.000	0.564	6.536	30.867	6.400	0.655	6.111	23.600	6.5333	0.6326	6.2192	48.567
N	3.000	0.518	2.388	199.700	3.000	0.513	2.395	218.900	3.000	0.512	2.416	210.867	3.000	0.517	2.392	186.300	3	0.5077	2.4124	230.57
O	7.967	0.826	7.641	0.833	7.433	0.920	7.333	11.733	8.800	0.675	8.490	3.567	7.933	0.831	7.596	1.533	7.3667	**0.929**	7.2584	10.567
P	9.867	0.713	9.420	45.533	9.967	0.688	9.577	71.233	10.133	0.655	9.835	51.533	9.800	**0.720**	9.383	53.967	9.9667	0.6884	9.5771	83.233
Q	15.633	0.941	15.338	14.200	16.600	0.843	16.283	137.933	17.133	0.792	16.807	10.200	15.500	**0.956**	15.263	7.800	16.333	0.8692	16.061	173.23
R	10.567	0.766	10.165	33.033	11.000	0.710	10.477	58.633	11.133	0.681	10.812	40.400	10.567	**0.767**	10.165	39.967	11	0.7103	10.462	100.67
Av.	7.600	0.682	7.189	51.811	7.670	0.671	7.289	81.361	7.922	0.632	7.521	53.878	7.569	0.685	7.159	52.880	7.656	0.672	7.275	85.376

Table 8
Results of the different constructive algorithms applied on set of instances JP2.

	SCH				BPGD				FF				PBP				TPS			
	N	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*
A	4.000	**0.626**	3.435	90.000	4.000	0.594	3.653	97.367	4.067	0.584	3.688	84.100	4.000	0.605	3.579	95.345	4	0.5926	3.6594	144.59
B	12.000	0.728	11.644	74.767	12.300	0.698	11.901	51.033	12.833	0.636	12.477	22.500	11.967	**0.731**	11.596	18.133	12.233	0.7014	11.848	48.467
C	7.700	0.665	7.217	134.633	7.767	0.653	7.260	105.933	7.933	0.612	7.452	79.467	7.517	**0.690**	7.094	79.828	7.7667	0.6522	7.2594	150.47
F	3.000	0.510	2.488	225.667	3.000	0.498	2.503	83.400	3.000	0.501	2.499	80.500	3.000	**0.511**	2.468	79.700	3	0.4921	2.5384	123
H	14.433	**0.730**	13.985	49.467	14.667	0.704	14.281	67.300	15.100	0.661	14.736	19.200	14.433	**0.730**	13.995	14.567	14.633	0.7091	14.219	88.167
L	4.200	0.563	3.883	93.600	4.133	0.564	3.886	67.967	4.167	0.556	3.914	59.667	4.167	**0.571**	3.823	71.933	4.1667	0.561	3.8764	70.2
M	6.633	0.627	6.264	94.767	6.833	0.594	6.354	78.200	6.967	0.564	6.524	63.667	6.533	**0.641**	6.194	68.067	6.8	0.6	6.328	85.467
O	8.300	0.769	7.857	21.700	8.733	0.691	8.429	34.600	9.200	0.618	8.888	11.567	8.233	**0.774**	7.843	5.533	8.6333	0.7038	8.3437	33.5
S	3.000	0.517	2.633	27.900	2.867	0.556	2.603	16.900	2.967	0.526	2.644	14.100	2.967	0.521	2.605	13.333	2.8667	**0.561**	2.5877	14.433
T	10.833	0.887	10.556	9.567	11.167	0.846	10.969	24.467	12.100	0.736	11.792	3.500	10.833	**0.890**	10.542	2.033	11.067	0.8599	10.883	24.333
U	6.133	0.732	5.830	19.167	6.200	0.736	5.901	18.833	6.933	0.571	6.550	8.500	6.100	0.737	5.826	7.600	6.1667	**0.744**	5.8871	17.133
V	5.233	0.942	5.115	2.833	5.033	0.990	5.027	0.633	5.267	0.936	5.194	0.467	5.200	0.951	5.091	0.600	5.0667	**0.982**	5.0395	0.4667
W	5.133	0.646	4.790	49.667	5.100	0.649	4.768	45.600	5.167	0.630	4.861	26.733	5.067	0.656	4.712	30.300	5.0333	**0.671**	4.7104	41.567
X	4.033	0.586	3.727	134.633	4.000	0.588	3.711	80.467	4.000	0.581	3.742	69.233	4.000	**0.593**	3.691	69.867	4.0345	0.5822	3.7171	78.103
Y	7.300	**0.720**	6.992	95.200	7.533	0.687	7.125	87.533	7.767	0.647	7.293	65.567	7.300	**0.720**	6.958	70.767	7.3667	0.7044	7.0639	91.1
Z	13.333	0.847	12.924	33.700	14.100	0.763	13.734	63.933	14.667	0.708	14.251	37.967	13.233	**0.856**	12.866	39.100	14.1	0.7633	13.735	64.433
Av.	7.204	0.693	6.834	72.329	7.340	0.676	7.007	57.760	7.633	0.629	7.282	40.421	7.159	0.699	6.805	41.669	7.308	0.680	6.981	67.214

In order to compare the improvement of each one of the local search procedures described in Section 5 we present the average improvement of each local search procedure over the solution obtained by the PBP algorithm. In Table 4 we present the average improvement of *F* obtained by both local search procedures LS1 and LS2 when applied to the solution obtained by the constructive algorithm PBP for each set of instances. In general there is a significant improvement with both local search strategies. We can observe that LS2 performs better than LS1, obtaining better average results in all sets of instances but one, with a similar computational effort. Tables 13 and 14 in Appendix A show the results obtained in each one of the classes on sets JP1 and JP2. The average results obtained on set JP1 by LS1 and LS2, 0.703 and 0.723, respectively, improve on the average solution quality from the best known algorithms results reported by Terashima-Marin et al. (2010), 0.690.

Note that the average improvement of LS2 on instances Nest-LB is much lower than the other sets of instances. This fact suggests that the impact of the local search depends on the ratio between the size of the pieces and the size of the bins. The average number of pieces per bin on instances sets Nest-SB, Nest-MB and Nest-LB is, respectively, 4.35, 8.19 and 14.34. Therefore, if the average number of pieces per bin is big enough then the local search procedure, which explores only the assignment of the pieces into the bin, is less effective. However, the local search produces relevant improvements on instances in which the average number of pieces per bin is lower. This confirms our conjecture that the assignment is more important when the number of pieces per bin is small.

Table 9
Results of the different constructive algorithms applied on each instance in Nest-SB.

	SCH				BPGD				FF				PBP				TPS			
	N	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*
albano	6	0.474	5.216	12	6	0.419	5.811	18	6	0.677	5.521	10	6	0.474	5.216	10	6	0.419	5.811	18
shapes2	20	0.300	19.727	3	20	0.300	19.727	4	20	0.542	19.727	1	20	0.300	19.727	3	20	0.300	19.727	5
trousers	5	0.671	4.909	202	6	0.519	5.247	232	5	0.831	4.909	195	5	0.671	4.909	181	5	0.677	4.886	185
shapes0	14	0.245	13.390	8	14	0.245	13.390	11	14	0.500	13.390	14	14	0.245	13.390	11	14	0.245	13.391	10
shapes1	13	0.272	12.714	22	13	0.272	12.714	22	13	0.528	12.714	19	13	0.272	12.714	21	14	0.241	13.712	19
shirts	14	0.571	13.839	135	16	0.462	15.490	115	16	0.676	15.629	97	14	0.571	13.839	131	16	0.466	15.490	169
dighe2	3	0.359	2.654	0	3	0.350	2.662	1	3	0.611	2.654	0	3	0.359	2.654	0	3	0.350	2.662	1
dighe1	3	0.406	2.771	3	3	0.406	2.771	3	3	0.677	2.817	4	3	0.406	2.771	3	3	0.406	2.772	2
fu	8	0.362	7.455	1	8	0.357	7.455	1	8	0.602	7.455	1	8	0.362	7.455	1	8	0.348	7.584	1
han	5	0.435	4.237	33	5	0.430	4.277	38	5	0.619	4.277	11	5	0.435	4.237	27	5	0.430	4.277	26
jakobs1	9	0.371	8.341	1	10	0.305	9.341	10	9	0.576	8.682	4	9	0.371	8.341	2	10	0.305	9.341	10
jakobs2	7	0.402	6.682	10	7	0.401	6.707	9	7	0.640	6.818	11	7	0.402	6.682	9	7	0.401	6.707	11
mao	4	0.432	3.706	22	4	0.430	3.831	27	4	0.702	3.706	16	4	0.432	3.706	19	4	0.428	3.706	20
poly1a	3	0.438	2.892	24	3	0.441	2.896	14	4	0.513	3.420	13	3	0.438	2.892	20	3	0.438	2.856	24
poly2a	7	0.363	6.420	52	6	0.433	5.977	47	7	0.575	6.420	50	7	0.363	6.420	49	7	0.360	6.280	53
poly3a	10	0.385	9.420	106	9	0.434	8.824	86	9	0.660	8.949	82	10	0.386	9.420	92	10	0.384	9.420	135
poly4a	13	0.381	12.515	206	12	0.436	11.860	147	12	0.657	11.997	116	13	0.386	12.439	160	13	0.385	12.439	242
poly5a	16	0.400	15.420	297	16	0.398	15.420	181	16	0.630	15.517	176	16	0.400	15.420	274	16	0.403	15.420	480
poly2b	7	0.409	6.632	31	7	0.395	6.843	27	7	0.652	6.629	39	7	0.410	6.506	33	7	0.409	6.615	70
poly3b	9	0.444	8.916	89	9	0.441	8.918	98	10	0.607	9.420	87	9	0.444	8.916	86	9	0.449	8.768	90
poly4b	12	0.430	11.210	151	12	0.414	11.420	91	12	0.645	11.420	136	12	0.430	11.210	147	12	0.419	11.420	235
poly5b	14	0.428	13.426	227	14	0.432	13.420	228	14	0.658	13.448	208	14	0.431	13.284	202	14	0.431	13.420	270
swimm	9	0.416	8.877	165	10	0.365	9.335	127	10	0.588	9.486	121	9	0.418	8.806	165	9	0.414	8.865	170
Av.	9.174	0.409	8.755	78.261	9.261	0.395	8.884	66.826	9.304	0.624	8.913	61.348	9.174	0.409	8.737	71.565	9.348	0.396	8.938	97.652

Table 10
Results of the different constructive algorithms applied on each instance in Nest-MB.

	SCH				BPGD				FF				PBP				TPS			
	N	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*
albano	3	0.480	2.873	26	3	0.480	2.873	25	3	0.491	2.667	24	3	0.480	2.873	22	3	0.480	2.895	36
shapes2	10	0.345	9.933	3	10	0.346	9.933	7	10	0.347	9.933	4	10	0.349	9.933	5	10	0.344	9.933	10
trousers	3	0.555	2.667	156	3	0.556	2.667	194	3	0.555	2.667	197	3	0.555	2.667	147	3	0.556	2.667	241
shapes0	7	0.271	6.762	71	7	0.271	6.762	58	7	0.287	6.524	36	7	0.271	6.762	60	7	0.271	6.762	66
shapes1	7	0.282	6.476	181	7	0.282	6.476	166	6	0.367	5.905	92	7	0.282	6.476	171	7	0.281	6.570	172
shirts	8	0.518	7.844	160	8	0.514	7.849	174	8	0.517	7.866	195	8	0.518	7.844	136	8	0.514	7.891	167
dighe2	1	0.823	0.952	8	1	0.823	0.952	11	1	0.823	0.952	14	1	0.823	0.952	9	1	0.823	0.951	9
dighe1	2	0.368	1.333	9	2	0.368	1.333	7	2	0.368	1.333	8	2	0.368	1.333	8	2	0.368	1.333	10
fu	4	0.443	3.571	1	4	0.443	3.571	1	4	0.454	3.429	1	4	0.443	3.571	1	4	0.443	3.571	2
han	3	0.387	2.203	32	3	0.387	2.203	27	3	0.376	2.290	34	3	0.387	2.203	27	3	0.391	2.290	27
jakobs1	4	0.570	3.333	54	4	0.567	3.250	52	4	0.552	3.417	53	4	0.570	3.333	45	4	0.567	3.250	52
jakobs2	4	0.401	3.250	36	4	0.401	3.250	35	4	0.412	3.333	40	4	0.401	3.250	36	4	0.401	3.251	36
mao	3	0.271	2.328	20	3	0.271	2.328	19	3	0.295	2.171	23	3	0.271	2.328	18	3	0.243	2.518	19
poly1a	2	0.308	1.536	36	2	0.300	1.598	25	2	0.308	1.536	39	2	0.308	1.536	35	2	0.300	1.597	25
poly2a	4	0.339	3.308	79	4	0.339	3.308	63	4	0.367	3.308	60	4	0.339	3.308	64	4	0.343	3.308	59
poly3a	5	0.419	4.705	184	5	0.426	4.632	137	5	0.414	4.790	146	5	0.419	4.665	165	5	0.424	4.682	137
poly4a	7	0.392	6.386	269	7	0.413	6.308	244	7	0.400	6.328	216	7	0.392	6.386	216	7	0.398	6.308	218
poly5a	8	0.443	7.850	441	8	0.447	7.667	362	8	0.445	7.958	387	8	0.449	7.905	370	8	0.446	7.950	339
poly2b	4	0.389	3.436	78	4	0.379	3.462	64	4	0.368	3.526	79	4	0.389	3.436	72	4	0.383	3.442	88
poly3b	5	0.434	4.582	148	5	0.430	4.564	115	5	0.433	4.581	133	5	0.432	4.582	170	5	0.441	4.507	158
poly4b	6	0.465	5.801	278	6	0.463	5.848	245	7	0.383	6.308	207	6	0.465	5.801	277	6	0.470	5.776	265
poly5b	7	0.478	6.936	386	7	0.476	6.826	488	7	0.477	6.933	439	7	0.478	6.936	368	7	0.476	6.826	488
swimm	5	0.397	4.891	311	5	0.401	4.870	227	5	0.392	4.907	365	5	0.397	4.891	238	5	0.401	4.870	232
Av.	4.870	0.425	4.476	129.000	4.870	0.425	4.458	119.391	4.870	0.427	4.464	121.391	4.870	0.425	4.477	115.652	4.870	0.425	4.485	124.174

The next set of computational experiments evaluate the impact of allowing free rotation of the pieces. In addition, we investigate the influence of the Initial Given rotation and how that impacts the solution quality. Our conjecture is that data sets are created with a good initial orientation with respect to packing and therefore keeping this orientation fixed would produce good solutions. For some applications i.e. garment manufacturing the pattern and bias of the material is important, and rotation angles are finite and specific. However, with many other materials there is no practical reason for any given initial rotation. Table 5 shows the results obtained for the Nest-MB instances by the PBP constructive algorithm with three configurations. In columns "Initial Given" the orientation of the pieces in the original data is kept and for each instance we consider the finite rotation angles given in Table 1. In columns "Initial Random" the orientation of the pieces in the original data is rotated a random angle and the permitted orientation of the pieces are considered starting from the new angle of rotation. The columns headed "Free Rot" present the results obtained when the pieces are allowed to rotate freely, regardless of their permitted orientations.

If we compare the results obtained by "Initial Given" and "Initial Random", we observe that "Initial Given" gets better solutions in seventeen of the twenty-three cases, and of those cases where "Initial Given" is worse, all but one arise in the *poly* data sets that are artificial instances with arbitrary shapes for which the initial orientation is not important. This suggests that the initial orienta-

Table 11
Results of the different constructive algorithms applied on each instance in Nest-LB.

	SCH				BPGD				FF				PBP				TPS			
	N	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*
albano	2	0.387	1.472	51	2	0.388	1.533	53	2	0.388	1.533	55	2	0.387	1.472	45	2	0.383	1.500	70
shapes2	5	0.439	4.800	25	5	0.441	4.800	27	5	0.449	4.700	23	5	0.452	4.700	41	5	0.439	4.799	22
trousers	2	0.445	1.500	504	2	0.437	1.500	635	2	0.435	1.500	578	2	0.445	1.500	433	2	0.450	1.500	480
shapes0	4	0.290	3.393	61	4	0.290	3.393	54	4	0.314	3.393	35	4	0.290	3.393	49	4	0.291	3.393	53
shapes1	4	0.302	3.286	134	4	0.302	3.286	108	4	0.317	3.214	86	4	0.302	3.286	103	4	0.285	3.428	86
shirts	4	0.644	3.926	554	4	0.640	3.935	708	4	0.642	3.928	501	4	0.644	3.926	456	5	0.498	4.269	715
dighe2	1	0.260	0.590	14	1	0.260	0.587	26	1	0.260	0.587	21	1	0.260	0.587	21	1	0.260	0.587	42
dighe1	1	0.329	0.835	43	1	0.329	0.835	37	1	0.329	0.835	35	1	0.329	0.835	35	1	0.329	0.833	19
fu	2	0.505	1.681	23	2	0.505	1.681	25	2	0.505	1.681	22	2	0.505	1.681	18	2	0.505	1.680	25
han	2	0.300	1.217	65	2	0.300	1.217	57	2	0.300	1.217	52	2	0.300	1.217	53	2	0.311	1.217	51
jakobs1	2	0.608	1.763	72	2	0.609	1.875	64	2	0.608	1.812	49	2	0.608	1.763	67	2	0.609	1.875	64
jakobs2	2	0.450	1.875	71	2	0.450	1.875	70	2	0.436	1.933	69	2	0.450	1.875	71	2	0.456	1.812	79
mao	2	0.231	1.346	29	2	0.231	1.346	32	2	0.231	1.346	29	2	0.231	1.346	30	2	0.213	1.319	35
poly1a	1	0.355	0.907	80	1	0.355	0.907	96	1	0.355	0.907	79	1	0.355	0.907	96	1	0.355	0.856	128
poly2a	2	0.369	1.723	194	2	0.371	1.744	233	2	0.366	1.711	151	2	0.369	1.723	198	2	0.366	1.770	299
poly3a	3	0.380	2.634	330	3	0.372	2.701	378	3	0.373	2.687	335	3	0.380	2.634	269	3	0.383	2.544	343
poly4a	4	0.382	3.521	682	4	0.385	3.481	581	4	0.384	3.500	511	4	0.382	3.521	538	4	0.377	3.588	666
poly5a	5	0.405	4.299	911	5	0.398	4.348	812	5	0.400	4.402	807	5	0.405	4.299	888	5	0.404	4.273	869
poly2b	2	0.442	1.898	217	2	0.445	1.917	239	2	0.439	1.925	236	2	0.442	1.898	223	2	0.448	1.873	180
poly3b	3	0.396	2.590	255	3	0.408	2.472	319	3	0.395	2.565	293	3	0.396	2.590	260	3	0.403	2.651	285
poly4b	4	0.389	3.265	467	4	0.398	3.269	682	4	0.400	3.231	593	4	0.389	3.255	474	4	0.393	3.270	689
poly5b	4	0.472	3.754	1010	4	0.464	3.828	1157	4	0.467	3.853	938	4	0.472	3.754	949	4	0.468	3.860	1063
swimm	3	0.364	2.593	912	3	0.375	2.557	797	3	0.356	2.657	1045	3	0.364	2.593	772	3	0.364	2.613	797
Av.	2.783	0.398	2.386	291.478	2.783	0.398	2.395	312.609	2.783	0.398	2.396	284.478	2.783	0.398	2.381	264.739	2.826	0.391	2.413	306.957

Table 12
Results of the different constructive algorithms applied on each instance in Jotika.

	SCH				BPGD				FF				PBP				TPS			
	N	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*	*N*	*F*	*K*	*T*
jotika40	8	0.612	7.378	23	8	0.616	7.385	19	8	0.628	7.299	38	8	0.612	7.378	25	8	0.616	7.385	47
jotika50	10	0.620	9.327	41	10	0.636	9.191	28	10	0.602	9.526	55	10	0.641	9.191	48	10	0.630	9.209	63
jotika60	11	0.659	10.299	43	11	0.648	10.432	57	11	0.625	10.598	88	11	0.668	10.194	53	11	0.642	10.466	76
jotika70	12	0.709	11.358	78	12	0.689	11.653	132	12	0.689	11.748	71	12	0.695	11.536	81	12	0.687	11.635	127
han80	10	0.711	9.286	139	10	0.697	9.422	104	10	0.687	9.523	172	10	0.711	9.286	136	10	0.694	9.485	247
han100	16	0.722	15.496	161	16	0.719	15.462	131	17	0.659	16.158	132	16	0.732	15.325	195	16	0.710	15.648	210
han120	16	0.749	15.670	174	16	0.751	15.555	283	17	0.692	16.127	314	16	0.749	15.679	259	17	0.682	16.243	313
han150	22	0.780	21.520	372	24	0.666	23.222	552	23	0.726	22.162	366	22	0.778	21.526	385	23	0.708	23.000	778
Av.	13.125	0.695	12.542	128.88	13.375	0.678	12.79	163.25	13.500	0.664	12.892	154.5	13.125	0.698	12.514	147.75	13.375	0.671	12.884	232.625

Table 13
Local search on JP1 instances.

	PBP	PBP-LS1	PBP-LS2	BKR
A	0.614	0.614	0.614	0.605
B	0.781	0.858	0.878	0.929
C	0.729	0.736	0.736	0.763
D	0.590	0.591	0.590	0.579
E	0.522	0.525	0.529	0.412
F	0.515	0.520	0.564	0.496
G	0.724	0.797	0.809	0.814
H	0.766	0.868	0.810	0.928
I	0.628	0.637	0.652	0.627
J	0.672	0.675	0.701	0.665
K	0.761	0.763	0.763	0.718
L	0.588	0.590	0.619	0.512
M	0.655	0.655	0.655	0.589
N	0.517	0.520	0.668	0.503
O	0.831	0.848	0.848	0.823
P	0.720	0.720	0.852	0.678
Q	0.956	0.965	0.965	1.000
R	0.767	0.767	0.767	0.771
Av.	0.685	0.703	0.723	0.690
Av. time	52	129	161	50
Impv.		1.74%	3.80%	

Table 14
Local search on JP2 instances.

	PBP	PBP-LS1	PBP-LS2
A	0.605	0.642	0.680
B	0.731	0.752	0.760
C	0.690	0.691	0.691
F	0.511	0.530	0.551
H	0.730	0.766	0.828
L	0.571	0.575	0.576
M	0.641	0.664	0.681
O	0.774	0.786	0.788
S	0.521	0.537	0.522
T	0.890	0.921	0.921
U	0.737	0.764	0.761
V	0.951	0.976	0.989
W	0.656	0.664	0.669
X	0.593	0.607	0.632
Y	0.720	0.725	0.726
Z	0.856	0.858	0.893
Av.	0.699	0.716	0.729
Av. time	41.669	136.163	123.344
Impv.		1.753	3.068

tion given to the pieces is advantageous in the majority of nesting instances coming from real problems, allowing good matchings between the pieces and the edges of the bin. In Fig. 8 we can observe that the initial angle of rotations on solution (b) is clearly better for packing, allowing the larger pieces to be packed together more effectively. In solution (a), where we randomized the initial angle of rotations, pieces have fewer matching edges leading to

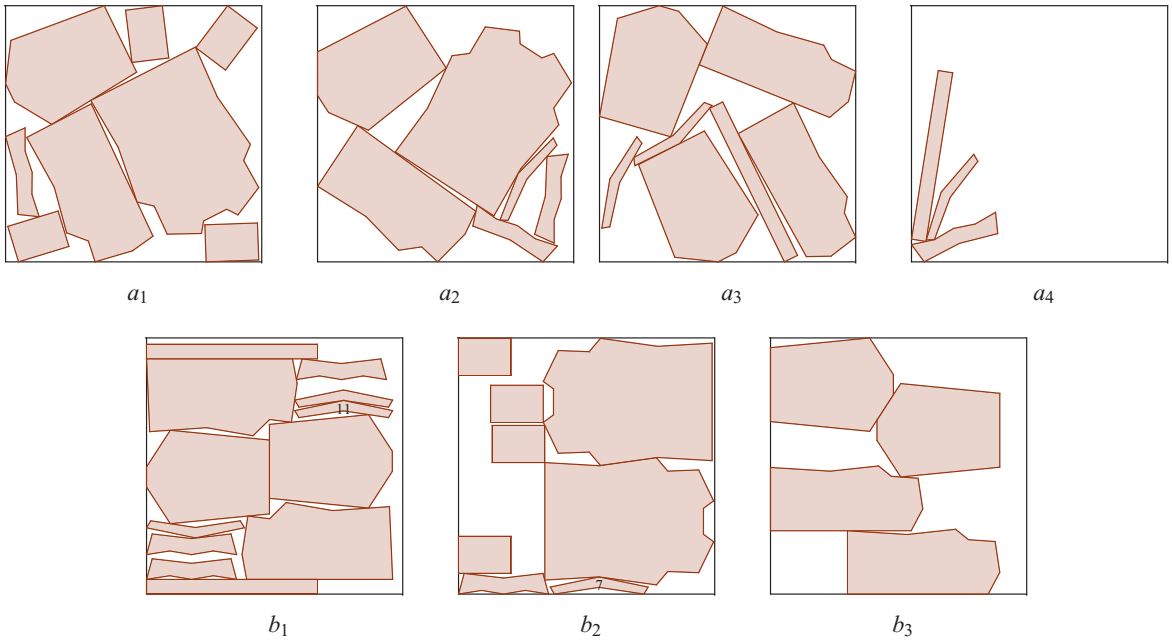


Fig. 8. Solution *a* with "Initial Random", needed 4 bins; solution *b* with "Initial Given", needed 3 bins.

a worse solution. A more significant impact can be found in instances *dighe1* and *dighe2*. These are jigsaw puzzle instances where only the initial orientation of the pieces allows the perfect fit.

Comparing these results with those obtained by "Free Rot", this variant found better result than both the finite rotation variants in fourteen cases. Note that our "get rotations" algorithm is not dependent on the initial angle of rotation given to the pieces. Hence, when there is no practical reason for setting specific angles of rotation, our free rotation approach removes any bias from how the data sets are created. Also note that on average the number of bins is better for "Free Rot", but its F value is inferior to that of "Initial Given". Although in most of the instance in which "Free Rot" does not obtain the best result the distance to best is very small, there are very large discrepancies on the two jigsaw instances, especially on *dighe2*, in which the algorithm with free rotation needs an extra bin, and as a result the F value is significantly lower.

In summary, when the instances come from real problems or are jigsaw instances they are given with an initial orientation which will produce the best or near-best solution. Our algorithm with free rotation, in which every time a piece is packed its orientation is chosen according to the partial configuration of the bin, is able to improve on the results obtained using this initial orientation for most of the instances and to get similar solutions for the remaining ones, except for jigsaw instances in which just one bad selection of the orientation of one piece can lead to much worse solutions. Apart from these very specific cases, the proposed algorithm, which is independent of the initial orientation of the pieces, tends to produce better solutions.

In Table 6 we present the results obtained in the three real instances proposed by BUTECH-PORCELANOSA. For these instances the objective is to reduce the total number of bins, N. For illustrative purposes the table also shows the average percentage utilization $U = \sum_{i=1}^{N} U_i/N$, as well as the computing times T. Concerning the algorithms, since the upper bounds on the number of bins and the number of pieces are very large, solving the corresponding IP models many times, as required by TPS, is computationally prohibitive. This variant of the algorithm is therefore not included. However, algorithm BPGD, which solves a similar IP formulation but far fewer times, produces a solution in reasonable computing time and is included. Nevertheless, the best algorithm in terms of computational time and quality of the solutions is, again, PBP, that always provides average utilizations greater than 90% and up to 97.49%. These solutions considerably improved the efficiency of previous manual solutions.

8. Conclusions

In this paper we have addressed a new bin packing problem with irregular pieces that can be freely rotated for which there is no specialized algorithm despite the fact that the problem arises in industry. It was motivated by the ceramic tile sector, from which we have tested the algorithm in several instances provided by BUTECH (PORCELANOSA group) in Spain. Due to the high cost of the material used, the main objective is to obtain high utilization solutions.

Besides the instances provided by the company we have used several sets of test instances, some of them previously published and other adapted from related problems, in order to have instance with very different characteristics. That has allowed us to study the behavior of the different strategies developed in this paper on a wide range of cases.

We have solved the problem by decomposing it into two phases, the assignment of pieces to bins and the packing of the assigned pieces in the corresponding bin. As one of the objectives of the study was to investigate the value of this separation strategy versus the assignment as a consequence of the packing decision, we have developed and tested several assignment procedures, including the simple sequential assignment method. The results show that a partial assignment is the most successful strategy. This strategy is known to work well in one dimensional and two-dimensional bin packing problems with rectangular pieces and it is also the best in this case of irregular pieces.

We have also investigated the relative impact of assignment and packing depending on the data instance and found it highly dependent on the average number of pieces per bin. The assignment is more important when the number of pieces per bin is small. In particular, the local searches developed to improve the solutions, based on modifying the assignment of pieces to bins, work better when only a few pieces fit into each bin.

Finally, we have studied the impact of the orientation of pieces given in the original data and how free rotation helps to improve the solutions by modifying the orientation of the pieces. Our results show that in some cases, such as jigsaw puzzle instances in which only using the initial orientation of the pieces is possible to obtain the perfect fit, allowing free rotation cannot improve on the solutions and more likely will produce worse solutions if some wrong decisions about the orientation are made in the constructive process. However, in many other cases, our algorithm, which is not dependent of the initial orientation, produces better results, finding better orientations for some pieces.

As future work, we plan to extend the ideas developed here to other bin packing problems with irregular pieces, such as the problems arizing from cutting leather, in which bins are neither identical nor rectangular, and can even have some defects, defining non-usable zones.

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Appendix

In this section we present with more detail the computational results presented and discussed in Section 7.

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